WTO Dispute Analysis Dataset Codebook

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Abstract

This codebook documents all datasets used in the WTO Dispute Settlement Body analysis project. The core dataset is a country-year panel covering 196 polities over 1995–2024 (5,880 observations, 77 variables), integrating six major data sources: World Development Indicators (WDI), Worldwide Governance Indicators (WGI), UN General Assembly Ideal Points, V-Dem v15, COW National Material Capabilities (NMC 6.0), and WTO dispute records. A forthcoming dyads dataset will add bilateral trade (CEPII BACI), preferential trade agreement depth (DESTA / World Bank DTA), and formal military alliances (ATOP 5.1). Systematic missing data by source is documented in Section 2.

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Panel Dataset: `country_meta_1995_2024.csv`

Panel Structure

Attribute	Value
Unit of analysis	Country-year
Observations	5,880
Countries / Polities	196
Year range	1995–2024 (30 years)
Primary key	`(country, year)`
Total variables	77

Table 1: Panel dataset overview.

The dataset integrates six major data sources into a balanced country-year panel covering all WTO-relevant polities (members, observers, and non-members that appear in WTO dispute records), plus the EU as a collective actor.

A. Identifiers

Variable	Definition	Unit / Range	Source
`country`	Country / polity name	String	User-defined
`iso3c`	ISO 3166-1 alpha-3 code	3-letter	WDI / User
`ccode`	COW numeric code; -1 if no match	Integer	COW
`stateabb`	COW 3-letter abbreviation	String	COW
`un_country_name`	Official UN name	String	UN / WDI
`year`	Calendar year	1995–2024	—
`COWcode`	COW code for V-Dem/NMC join	Integer	COW / V-Dem

Table 2: Identifier variables.

B. WTO Membership & Participation

Source: WTO official membership records and `wto_cases.csv` (626 cases, 1995–2024).

Variable	Definition	Range	Notes
wto_member	Member in that year	Binary (0/1)	112 (1995) to 166 (2024)
wto_accession_year	Year of WTO accession	1995–2024	NaN for non-members
complainant	Filed ≥1 dispute that year	Binary (0/1)	—
respondent	Faced ≥1 dispute that year	Binary (0/1)	—
third_party	Joined as third party that year	Binary (0/1)	—
wto_participant	Any dispute activity	Binary (0/1)	OR of above three
cum_complainant	Cumulative disputes filed	0–25	US and EU both reach 25
cum_respondent	Cumulative disputes faced	0–29	US reaches 29
cum_third_party	Cumulative third-party appearances	0–30	US reaches 30

Table 3: WTO membership and dispute participation variables.

C. EU / Eurozone Membership

Variable	Definition	Values	Notes
eu_member	EU member in that year	Binary (0/1)	28 unique countries ever
euro_join_year	Year of Euro adoption	1999–2023	NaN: 89.8% of obs
euro_member	Euro user in that year	Binary (0/1)	Derived from above

Table 4: EU and Eurozone membership variables. Source: EU official records; ECB Eurozone records.

D. UN Voting Ideal Points

Primary citation: Bailey, Michael A., Anton Strezhnev, and Erik Voeten. 2017. "Estimating Dynamic State Preferences from United Nations Voting Data." *Journal of Conflict Resolution* 61(2): 430–456.

Data version: Voeten, Erik. July 28, 2025 (Georgetown University). Raw voting data: Fjelstul, Joshua, Simon Hug, and Christopher Kilby. 2025. *The Review of International Organizations*. Dataset: https://unvotes.unige.ch/

Ideal points are estimated using a Bayesian IRT model on UNGA voting records. Higher values indicate alignment with the United States; lower values indicate alignment with Russia/China.

D.1 Ideal Point Estimates

Variable	Definition	Range
`idealpointfp`	Ideal point – final-passage votes only (preferred for most analyses)	−2.10 to +3.03
`idealpointall`	Ideal point – all votes incl. paragraphs and amendments; may over-weight specific issues (e.g., Gaza)	−2.36 to +3.06
`idealpointlegacy`	Legacy session-based estimation; $r = 0.988$ with `idealpointall`	−2.17 to +3.22

Table 5: UN Voting ideal point estimates.

D.2 Vote Counts and Posterior Quantiles

Variable	Definition	Range
`nvotesfp`	Number of final-passage votes used for FP estimate	1–109
`nvotesall`	Number of all votes used for ALL estimate	1–232
`nvotesLegacy`	Number of votes for legacy estimate	1–96
`q5fp`–`q100fp`	5th–100th percentile of MCMC posterior (FP model)	≈−2.4 to +3.7
`q5all`–`q100all`	5th–100th percentile of MCMC posterior (All model)	≈−2.6 to +3.5

Table 6: Vote counts and posterior uncertainty quantiles. `qofp` / `qoall` = posterior mean (equivalent to point estimates above).

E. V-Dem Democracy Indicators

Citation: Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, et al. 2025. "V-Dem Codebook v15." Varieties of Democracy (V-Dem) Project, University of Gothenburg.

Variable	Definition	Range	Codebook ref.
v2x_polyarchy	Electoral Democracy Index: freedom of expression, clean elections, suffrage, elected officials	$[0, 1]$	v15, p. 37
v2x_libdem	Liberal Democracy Index: electoral democracy plus minority rights protection	$[0, 1]$	v15, p. 38
v2x_regime	Regime type: 0 = Closed Autocracy; 1 = Electoral Autocracy; 2 = Electoral Democracy; 3 = Liberal Democracy	0–3	v15, p. 291
v2eltype_0	Legislative election held	0/1/NaN	Election years only
v2eltype_1	Presidential election held	0/1/NaN	Election years only
v2eltype_6	Constituent assembly election	0/1/NaN	Election years only
v2eltype_7	Other executive election	0/1/NaN	Election years only
is_leg_elec	Any legislative election (derived)	Binary (0/1)	—
is_pres_elec	Any presidential election (derived)	Binary (0/1)	—
election_binary	Any election, legislative or presidential	Binary (0/1)	—

Table 7: V-Dem indices and election variables. NaN in v2eltype_* denotes a non-election year, not missing data.

F. World Development Indicators (WDI) – Economic

Source: World Bank, World Development Indicators (WDI), accessed 2025. https://databank.worldbank.org/source/world-development-indicators

Taiwan supplements: GDP/GDPpc from DGBAS (Executive Yuan, Taiwan); FDI from UNCTAD Global Investment Data.

Unit standardization: Monetary values ÷ 1,000,000 → **Million USD**. WDI percentage values ÷ 100 → **Ratio (0–1)**.

Variable	Definition (WDI indicator code)	Unit	Range
gdp	Real GDP, constant 2015 USD (NY.GDP.MKTP.KD)	M USD	23.6 – 22,568,000
gdppc	Real GDP per capita, constant 2015 USD (NY.GDP.PCAP.KD)	USD	211 – 247,170
pop	Total population (SP.POP.TOTL)	Count	9,280 – 1.45B
gdp_growth_rate	Annual GDP growth rate (NY.GDP.MKTP.KD.ZG)	Ratio	−0.50 – +1.50
trade	(Exports + Imports) / GDP (NE.TRD.GNFS.ZS)	Ratio	0.0002 – 4.37
fdi	Net FDI inflows / GDP (BX.KLT.DINV.WD.GD.ZS)	Ratio	−1303 – +1283
fdi_inflow_usd	Net FDI inflows, absolute (BX.KLT.DINV.CD.WD)	M USD	−343k – +734k
exp_share	Exports of goods and services / GDP (NE.EXP.GNFS.ZS)	Ratio	0.000054 – 2.29
imp_share	Imports of goods and services / GDP (NE.IMP.GNFS.ZS)	Ratio	0.000156 – 2.08
unemployment_rate	Unemployment rate, national estimate (SL.UEM.TOTL.NE.ZS)	Ratio	0.00039 – 0.39

Table 8: WDI economic variables. fdi ratio can exceed ±100% for small open economies. Full FDI range: $-343,400$ to $+733,830$ M USD.

G. Worldwide Governance Indicators (WGI)

Citation: Kaufmann, Daniel, Aart Kraay, and Massimo Mastruzzi. 2010. "The Worldwide Governance Indicators: Methodology and Analytical Issues." *World Bank Policy Research Working Paper* No. 5430. https://info.worldbank.org/governance/wgi/

All six indicators are on a standard normal scale (−2.5 to +2.5); higher = better governance.

Variable	WGI Code	Definition	Range
`voice`	VA.EST	Voice and Accountability: citizen participation in selecting government; freedom of expression, association, and press	−2.31 to +1.80
`stability`	PV.EST	Political Stability and Absence of Violence/Terrorism	−3.18 to +1.76
`efficiency`	GE.EST	Government Effectiveness: quality of public services, civil service, and policy formulation	−2.44 to +2.47
`reg_quality`	RQ.EST	Regulatory Quality: formulating policies that promote private sector development	−2.53 to +2.31
`law`	RL.EST	Rule of Law: contract enforcement, property rights, courts, police	−2.33 to +2.12
`corruption`	CC.EST	Control of Corruption: extent to which public power is exercised for private gain	−1.97 to +2.46

Table 9: WGI governance indicators (higher = better). Pre-2002 gaps interpolated: 1995 backfilled from 1996; 1997, 1999, 2001 linearly interpolated.

H. Geographic & Classification Variables

Variable	Definition	Values / Unit
`region`	World Bank geographic region	7 categories: East Asia & Pacific; Europe & Central Asia; Latin America & Caribbean; Middle East & North Africa; North America; South Asia; Sub-Saharan Africa. `NaN` for EU.
`longitude`	Geographic centroid longitude	Decimal degrees; −175.2 to +179.1
`latitude`	Geographic centroid latitude	Decimal degrees; −41.3 to +121.0
`income`	World Bank income group	Low / Lower-middle / Upper-middle / High / Not classified. `NaN` for EU.

Table 10: Geographic and income classification. Source: WDI Extra fields; user-defined for Taiwan and EU.

I. Derived Variables

Variable	Definition	Formula	Range
log_gdppc	Natural log of real GDP per capita	ln(gdppc)	5.35 – 12.42
log_pop	Natural log of total population	ln(pop)	9.14 – 21.10

Table 11: Derived variables constructed within the project; no external source.

J. COW National Material Capabilities (NMC)

Citation: Singer, J. David, Stuart Bremer, and John Stuckey. 1972. "Capability Distribution, Uncertainty, and Major Power War, 1820–1965." In *Peace, War and Numbers*, ed. Bruce Russett. Beverly Hills: Sage.

Dataset: NMC 6.0 (1816–2016). Hosted by J. Michael Greig and Andrew J. Enterline, University of North Texas. Updated June 28, 2021. Codebook: COW/NMC_Documentation_v6_0_final_v2.pdf

Critical coverage note: NMC 6.0 ends in **2016**. All 196 countries have NaN for all NMC variables for 2017–2024 (1,568 missing observations total).

Var.	Definition	Unit	Range / Notes
cinc	Composite Index of National Capability: unweighted average of six capability shares. Codebook p. 4: "CINC is the average of a state's share of the system total for each of the six capability components."	Proportion (0–1)	2.4×10^{-7} – 0.231
milex	Military expenditures	Thousands USD (post-1914); thousands GBP (pre-1914)	-9 to $693,600,000$
milper	Military personnel	Thousands of persons	-9 to $2,930$
pec	Primary energy consumption	Thousands coal-ton equivalents	-9 to $5,900,300$
tpop	Total population (COW measure)	Thousands of persons	9 – $1,403,500$
upop	Urban population (cities > 100k; > 300k after 2002)	Thousands of persons	0 – $612,930$

Table 12: COW NMC variables. Raw sentinel -9 = missing; recode to NaN before modelling. All values are NaN for 2017–2024.

Systematic Missing Data

UN Voting Ideal Points

Polity (all 30 years)	Reason for exclusion
Taiwan	Not a UN member (PRC replaced ROC in 1971); no UNGA voting record
Hong Kong	Not a sovereign state; votes subsumed under China (PRC)
Macao, China	Not a sovereign state; votes subsumed under China (PRC)
EU	Not a UN member state; EU participates as observer only

Table 13: Polities entirely missing UN Voting ideal points across all years.

Partially missing (some years, due to non-membership or absent data): Afghanistan, Bosnia & Herzegovina, Dominican Republic, Gambia, Guinea-Bissau, Iraq, Kiribati, Kyrgyzstan, Liberia, Mauritania, Montenegro, Nauru, Niger, Palau, Sao Tome and Principe, Seychelles, Somalia, South Sudan, Switzerland (observer until 2002), Tajikistan, Timor-Leste, Tonga, Tuvalu, Uzbekistan, Vanuatu, Venezuela.

V-Dem Democracy Indices

Polities entirely absent from V-Dem v15 (all 30 years): EU, Hong Kong, Macao, Andorra, Monaco, San Marino, Liechtenstein, Antigua and Barbuda, Bahamas, Belize, Brunei Darussalam, Dominica, Grenada, Kiribati, Marshall Islands, Micronesia (Fed. States), Montenegro, Nauru, Palau, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Samoa, South Sudan, Tonga, Tuvalu.

COW National Material Capabilities

- **2017–2024 (all 196 countries):** NMC 6.0 ends in 2016.
- **Additional historical gaps:** EU, Hong Kong, Macao, micro-states (Andorra, Monaco, San Marino, Liechtenstein), South Sudan (pre-2011), Montenegro (pre-2006).

World Development Indicators (WDI)

Polity	Reason
North Korea	No WDI reporting; data isolation and international sanctions
Syria	Post-2011 conflict severely disrupted statistical capacity
Somalia	Prolonged state fragility; limited statistical infrastructure
South Sudan	Newly independent (2011); early years lack data
Lebanon	Severe economic crisis (2019–) disrupted reporting
EU	WDI covers member states; not the supranational entity
Liechtenstein, Monaco, San Marino	Small states not covered by WDI
Taiwan	Not a WDI member; supplemented from DGBAS

Table 14: Countries with substantial WDI gaps. `unemployment_rate` missing for 42.8% of obs (ILO non-reporting by developing countries). `trade`, `exp_share`, `imp_share` missing for 18.1%.

WGI Governance Indicators

Missing ≈5–9% of observations. Systematic gaps:

- **North Korea:** Not covered.
- **Small island states** (Kiribati, Nauru, Tuvalu, Palau, etc.): Partial coverage only.
- **EU:** Supranational entity; not applicable.
- **Pre-2002 years:** Interpolated (see Section 3).

Data Processing & Construction Notes

Unit Standardization

Transformation	Details
GDP / FDI monetary values	Divided by 1,000,000 → Million USD
WDI percentage variables	Divided by 100 → Ratio (0–1); applies to `trade`, `fdi`, `exp_share`, `imp_share`, `unemployment_rate`, `gdp_growth_rate`
COW NMC sentinel values	Raw −9 retained as-is in the dataset; must be recoded to `NaN` before modelling

Table 15: Unit standardization applied during data construction.

WGI Interpolation for Missing Years

WGI surveys were not published annually before 2002:

- **1995:** Backward fill using 1996 values.
- **1997:** Linear interpolation of 1996 and 1998 values.
- **1999:** Linear interpolation of 1998 and 2000 values.
- **2001:** Linear interpolation of 2000 and 2002 values.

WTO Dispute Analysis Results

Case Dataset (`wto_cases.csv`)

Dispute Stage	Cases	Share
Consultation	237	37.8%
Panel	175	27.9%
Appellate Body	115	18.4%
Mutually Agreed Solution	69	11.0%
Implementation & Compliance	23	3.7%
Retaliation & Arbitration	4	0.6%
Unknown	3	0.5%
Total	**626**	

Table 16: Dispute stage distribution for 626 WTO cases (DS1–DS626).

Top Litigants (Cumulative through 2024)

Country	As Complainant	As Respondent
U.S.	120	159
EU	110	89
Canada	38	23
Brazil	33	17
Japan	27	16
China	25	49
India	22	32
Argentina	22	22
Mexico	21	—
South Korea	20	19

Table 17: Top 10 complainants and respondents (1995–2024). "—" = not in top 10 for that role.

Document Processing Statistics

Metric	Value
Total PDFs processed	9,480
Successfully classified	9,398 (99.1%)
Manual review required	82
Scanned (OCR required)	78
Non-English	3
Other errors	1
Document types (consolidated)	35
Date coverage	≈90%

Table 18: Document processing pipeline statistics (February 2026).

Annual Network Metrics (Selected Years)

Year	Nodes	Edges	Conflict Density	Modularity	Comm.
1995	26	100	0.086	+0.196	3
2000	36	267	0.065	+0.019	3
2005	26	100	0.037	+0.182	4
2010	33	217	0.036	+0.031	2
2015	32	227	0.026	−0.052	3
2018	50	1,117	0.032	+0.086	4
2020	23	87	0.020	+0.173	2
2024	20	39	0.026	+0.200	1

Table 19: Selected annual network statistics from `wto_analysis_results_1995-2024.json`. Nodes = countries active in disputes that year; Comm. = communities (Louvain algorithm).

Dyads Dataset (Forthcoming)

The dyads dataset will be a **directed country-pair-year** panel covering all WTO-relevant country pairs, 1995–2024.

Planned file: `dyads_1995_2024.csv`
Unit: Directed dyad-year ($A \to B$, t)
Estimated rows: ≈600,000 (196×195 directed pairs $\times$ 30 years, filtered)

Source 1 — CEPII BACI (Bilateral Trade Flows)

Citation: Gaulier, Guillaume, and Soledad Zignago. 2010. "BACI: International Trade Database at the Product-Level. The 1994–2007 Version." *CEPII Working Paper* No. 2010-23. Centre d'Études Prospectives et d'Informations Internationales.

Access: http://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=37

BACI reconciles UN Comtrade importer and exporter reports using a CIF/FOB harmonization procedure. It is the standard source for bilateral product-level trade flows used in gravity models.

Raw BACI Structure (exporter × importer × HS6 × year)

Variable	Definition	Unit
`t`	Year	Integer
`i`	Exporter country (BACI/ISO numeric code)	Integer
`j`	Importer country (BACI/ISO numeric code)	Integer
`k`	HS 6-digit product code	String
`v`	Trade value	Thousands USD
`q`	Quantity	Metric tons

Table 20: Raw BACI file structure (one row per exporter–importer–product–year combination).

Planned Aggregation to Dyad-Year Level

Variable	Definition	Formula
`export_value_AB`	Total exports from A to B in year t	$\sum_k v_{ABkt}$
`import_value_AB`	Total imports A receives from B in year t	= `export_value_BA`
`bilateral_trade`	Total bilateral trade (both directions)	`export` + `import`
`log_trade`	Log-transformed bilateral trade	ln(`bilateral_trade`+ 1)

Table 21: Constructed dyad-year trade variables aggregated from BACI product-level data.

Systematic gaps: Taiwan is coded as "Taipei, Chinese" (numeric code 490) in Comtrade and requires manual remapping. Hong Kong and Macao may double-count entrepôt re-exports. North Korea data are unreliable or entirely absent.

Source 2 — ATOP v5.1 (Formal Military Alliances)

Citation: Leeds, Brett Ashley, Jeffrey M. Ritter, Sara McLaughlin Mitchell, and Andrew G. Long. 2002. "Alliance Treaty Obligations and Provisions, 1815–1944." *International Interactions* 28: 237–260.

Codebook: Leeds, Brett Ashley. 2022. *ATOP Codebook v5.1.* Rice University. http://www.atopdata.org

Coverage: 765 alliances, 1815–2018. All 2019–2024 dyad-years are `NaN` for ATOP variables.

Variable	Definition	Values
`dyad`	COW dyad ID (`mem1` $\times 1000 +$ `mem2`)	Integer
`mem1`, `mem2`	COW numeric country codes for the two states	Integer
`year`	Calendar year	1815–2018
`atopally`	Any alliance relationship in this dyad	Binary (0/1)
`defense`	Mutual defense obligation	Binary (0/1)
`offense`	Offensive military obligation	Binary (0/1)
`neutral`	Neutrality pledge	Binary (0/1)
`nonagg`	Non-aggression pact	Binary (0/1)
`consul`	Consultation / coordination obligation	Binary (0/1)
`bilatno`	Number of bilateral alliances in dyad	Integer
`multino`	Number of multilateral alliances in dyad	Integer
`asymm`	Any asymmetric obligation present	Binary (0/1)
`atopid1`–`atopid9`	ATOP IDs of alliances covering this dyad (up to 9)	Integer

Table 22: Key variables from `atop5_1dy.csv` (136,648 rows; undirected dyad-year). Merge to node panel via `ccode_A`/`ccode_B`. Non-COW polities (EU, Taiwan, HK, micro-states) have no ATOP match.

Source 3 — DESTA & World Bank DTA (PTA Depth)

DESTA

Citation: Dür, Andreas, Leonardo Baccini, and Manfred Elsig. 2014. "The Design of International Trade Agreements: Introducing a New Dataset." *The Review of International Organizations* 9(3): 353–375. https://www.designoftradeagreements.org/

File available: `Data/desta_dyadic.csv` (781 agreements, 1948–2023)

Variable	Definition	Values
`number`	Unique DESTA agreement identifier	Integer
`name`	Agreement name	String
`year`	Signing year	1948–2023
`entryforceyear`	Year agreement entered into force	Float
`wto_listed`	Notified to WTO	Binary (0/1)
`mar_typedepth`	**PTA type / depth** (ordinal): 1 = Partial scope; 2 = FTA; 3 = Customs union; 4 = Common market; 5 = Economic union	1–5
`mar_goods_nt`	National treatment clause for goods	Binary (0/1)
`mar_goods_mfn`	Most-favoured-nation clause for goods	Binary (0/1)
`ser_general`	General services liberalization provision	Binary (0/1)

Table 23: Key DESTA variables. File is agreement-level (781 rows) and must be expanded to all member-pair combinations before merging on `(iso3c_A, iso3c_B, year` $\geq$ `entryforceyear)`.

World Bank Deep Trade Agreements (DTA)

Citation: Mattoo, Aaditya, Nadia Rocha, and Michele Ruta, eds. 2020. *The Evolution of Deep Trade Agreements.* Washington, DC: World Bank. https://datatopics.worldbank.org/dta/

The World Bank DTA database extends DESTA with a continuous **Rasch-scaled depth index** derived from item response theory, enabling cross-agreement comparisons on a single scale.

Planned Variable	Definition	Source
`pta_dummy`	Any PTA in force between the dyad in year t	DESTA / WB DTA
`pta_depth_type`	Ordinal PTA depth (`mar_typedepth`, 1–5)	DESTA
`pta_depth_rasch`	Continuous Rasch-scaled depth index	World Bank DTA
`desta_number`	DESTA agreement ID (for traceability)	DESTA

Table 24: Planned dyad-year PTA variables. Systematic gaps: Taiwan's PTAs only partially coded in DESTA. DESTA v2 ends in 2023.

Full Dyad-Year Variable List (Planned)

Group	Planned Variables	Source
Identifiers	`iso3c_A`, `iso3c_B`, `year`, `ccode_A`, `ccode_B`, `dyad_id`	—
WTO Dispute	`dispute_AB`, `dispute_BA`, `third_party_AB`, `cum_disputes_AB`	`wto_cases.csv`
Bilateral Trade	`export_value_AB`, `bilateral_trade`, `log_trade`	CEPII BACI
Trade Agreement	`pta_dummy`, `pta_depth_type`, `pta_depth_rasch`	DESTA / WB DTA
Military Alliance	`atop_ally`, `atop_defense`, `atop_nonagg`, `atop_consul`	ATOP v5.1
UN Voting Alignment	`ideal_distance_fp`, `s_score`	Voeten 2025
Bilateral GDP	`log_gdp_product`, `log_gdp_ratio`	WDI (node panel)
Contiguity / Geography	`contiguous`, `log_distance`	COW / CEPII GeoDist

Table 25: Planned dyad-year variable groups for `dyads_1995_2024.csv`.